

☒ Easy Level (10 Questions)

Q1. What does SUM(column) do?

- A. Adds all numeric values in column
- B. Counts rows
- C. Finds maximum
- D. Calculates average

Answer: A

Explanation: SUM adds all numeric values.

Q2. Which function counts the number of rows?

- A. SUM()
- B. AVG()
- C. COUNT()
- D. MAX()

Answer: C

Explanation: COUNT() returns number of rows.

Q3. What does AVG(column) return?

- A. Minimum value
- B. Total sum
- C. Average value
- D. Number of rows

Answer: C

Explanation: AVG returns average.

Q4. Which clause groups rows?

- A. ORDER BY
- B. GROUP BY
- C. HAVING
- D. WHERE

Answer: B

Explanation: GROUP BY groups rows.

Q5. Which clause filters groups?

- A. WHERE
- B. HAVING
- C. ORDER BY
- D. FROM

Answer: B

Explanation: HAVING filters aggregated groups.

Q6. What is a subquery?

- A. A query inside another query
- B. A function
- C. A type of index

D. A table

Answer: A

Explanation: Subquery is a nested query.

Q7. What will this query return?

sql

CopyEdit

```
SELECT COUNT(*) FROM users;
```

A. Sum of users

B. Number of rows in users

C. Maximum ID

D. Average age

Answer: B

Explanation: Counts rows in users.

Q8. Can HAVING be used without GROUP BY?

A. Yes

B. No

Answer: A (Some DBMS allow it)

Explanation: Usually HAVING filters groups, but some DBMS allow without GROUP BY.

Q9. What is the result of this query?

sql

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SELECT department FROM employees GROUP BY department;

A. One row per department

B. All rows

C. One row per employee

D. Error

Answer: A

Explanation: Groups employees by department.

Q10. Which aggregate function ignores NULL values?

- A. SUM()
- B. COUNT(column)
- C. AVG()
- D. All of the above

Answer: D

Explanation: Aggregate functions ignore NULLs in calculation.

📄 Medium Level (20 Questions)

Q11. What does this query do?

sql

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```
SELECT department, AVG(salary) FROM employees GROUP BY department;
```

- A. Averages salary across all employees
- B. Averages salary per department

C. Counts employees

D. Filters employees

Answer: B

Explanation: Groups by department and averages salary per group.

Q12. What's the difference between WHERE and HAVING?

A. WHERE filters before grouping, HAVING filters after

B. No difference

C. HAVING filters rows, WHERE filters groups

D. HAVING is faster

Answer: A

Explanation: WHERE filters rows before aggregation; HAVING filters groups after.

Q13. Which of the following is a valid subquery?

A.

sql

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SELECT * FROM (SELECT name FROM students) AS temp;

B.

sql

CopyEdit

SELECT name FROM students WHERE age > (SELECT AVG(age) FROM students);

C. Both

D. None

Answer: C

Explanation: Both are valid subqueries.

Q14. What does this query do?

sql

CopyEdit

```
SELECT department FROM employees GROUP BY department HAVING COUNT(*) > 5;
```

- A. Lists all departments
- B. Departments with more than 5 employees
- C. Errors out
- D. Counts employees only

Answer: B

Explanation: Filters departments having more than 5 employees.

Q15. What is the result of:

```
sql
```

CopyEdit

```
SELECT COUNT(salary) FROM employees;
```


- A. Number of non-null salary rows
- B. Number of all rows
- C. Sum of salaries
- D. Average salary

Answer: A

Explanation: Counts non-null values only.

Q16. Which aggregate function returns the maximum value?

- A. MIN()
- B. MAX()
- C. SUM()
- D. AVG()

Answer: B

Explanation: MAX returns the highest value.

Q17. What does this query return?

sql

CopyEdit

SELECT name FROM employees WHERE salary > (SELECT AVG(salary) FROM employees);

- A. Employees earning below average
- B. Employees earning above average
- C. All employees
- D. Error

Answer: B

Explanation: Filters employees with salary above average.

Q18. Can subqueries return multiple rows?

- A. Yes
- B. No

Answer: A

Explanation: Subqueries can return single or multiple rows depending on context.

Q19. What will this return?

sql

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SELECT department, COUNT(*) FROM employees GROUP BY department ORDER BY COUNT(*) DESC;

- A. Departments sorted by employee count descending
- B. Departments sorted alphabetically
- C. Random order
- D. Syntax error

Answer: A

Explanation: Orders departments by employee count descending.

Q20. What happens if you use WHERE with aggregate functions?

- A. Works fine
- B. Syntax error
- C. Filters on aggregate results
- D. Filters groups

Answer: B

Explanation: WHERE cannot filter aggregated results; HAVING must be used.

☒ **Medium Level (Continued: Q21–Q30)**

Q21. What does this query do?

sql

CopyEdit

```
SELECT department, SUM(salary) AS total_salary FROM employees GROUP BY department HAVING SUM(salary) > 100000;
```

- A. Lists departments with total salary over 100000
- B. Lists all departments
- C. Lists departments with salary under 100000
- D. Syntax error

Answer: A

Explanation: HAVING filters groups with total salary exceeding 100000.

Q22. Which aggregate function counts all rows, including NULLs?

- A. COUNT(column_name)
- B. COUNT(*)
- C. COUNT(DISTINCT column)
- D. SUM(column)

Answer: B

Explanation: COUNT(*) counts all rows regardless of NULLs.

Q23. What will happen if a subquery returns multiple rows where a single value is expected?

- A. Query executes successfully
- B. Error: subquery returns more than one row
- C. Returns first value only
- D. Returns NULL

Answer: B

Explanation: When a subquery expecting a scalar returns multiple rows, an error occurs.

Q24. Which clause can be used with GROUP BY to filter aggregated data?

- A. WHERE
- B. HAVING
- C. ORDER BY
- D. FROM

Answer: B

Explanation: HAVING filters after aggregation.

Q25. What will this query return?

sql

CopyEdit

SELECT MAX(salary) FROM employees WHERE department = 'IT';

- A. Max salary overall
- B. Max salary in IT department
- C. Average salary in IT
- D. Min salary in IT

Answer: B

Explanation: MAX applied on filtered rows by WHERE.

Q26. Which SQL keyword allows you to assign an alias to a subquery?

- A. AS
- B. ALIAS
- C. RENAME
- D. WITH

Answer: A

Explanation: AS is used to assign alias.

Q27. What is the output of this query?

sql

CopyEdit

```
SELECT department, AVG(salary) FROM employees GROUP BY department HAVING AVG(salary) > 50000;
```

- A. Departments with average salary above 50,000
- B. Departments with average salary below 50,000
- C. All departments
- D. Error

Answer: A

Explanation: HAVING filters groups based on AVG.

Q28. What is a correlated subquery?

- A. A subquery independent of outer query
- B. A subquery that references columns from outer query
- C. Subquery with HAVING clause
- D. Subquery inside FROM clause only

Answer: B

Explanation: Correlated subqueries depend on outer query values.

Q29. Which aggregate function will ignore NULL values when calculating?

- A. SUM()
- B. COUNT(column)
- C. AVG()
- D. All of the above

Answer: D

Explanation: Aggregate functions ignore NULLs.

Q30. How can you find the total number of employees per department including departments with zero employees?

- A. INNER JOIN employees and departments
- B. LEFT JOIN departments and employees
- C. Use WHERE clause
- D. Use subquery only

Answer: B

Explanation: LEFT JOIN includes all departments even without employees.

☒ Hard Level (10 Questions)

Q31. What is the output of this correlated subquery?

sql

CopyEdit

```
SELECT e1.name, e1.salary FROM employees e1 WHERE e1.salary > (SELECT AVG(e2.salary) FROM employees e2 WHERE e2.department = e1.department);
```

- A. Employees with salary above overall average
- B. Employees with salary above average in their own department
- C. Error due to correlation
- D. Employees below average

Answer: B

Explanation: Correlated subquery compares salary with dept average.

Q32. What does this query accomplish?

sql

CopyEdit

SELECT department, COUNT(DISTINCT employee_id) FROM employees GROUP BY department HAVING COUNT(DISTINCT employee_id) > 10;

- A. Counts total employees
- B. Counts unique employees per department with more than 10 employees
- C. Filters employees under 10
- D. Syntax error

Answer: B

Explanation: Counts unique employees per dept and filters groups.

Q33. Which query retrieves departments without employees?

A.

sql

CopyEdit

SELECT department FROM departments WHERE id NOT IN (SELECT department FROM employees);

B.

sql

CopyEdit

```
SELECT department FROM employees WHERE department IS NULL;
```

C.

sql

CopyEdit

```
SELECT department FROM departments LEFT JOIN employees ON departments.id = employees.department WHERE employees.id IS NULL;
```

D. Both A and C

Answer: D

Explanation: Both queries retrieve departments without employees.

Q34. What will this query result in?

sql

CopyEdit

SELECT name FROM employees WHERE salary > ALL (SELECT salary FROM employees WHERE department = 'HR');

- A. Names of employees earning more than everyone in HR
- B. Names of employees in HR
- C. Syntax error
- D. Average salary

Answer: A

Explanation: > ALL compares with all salaries in HR.

Q35. What is true about subqueries in the FROM clause?

- A. They can be aliased
- B. They can be used as tables

C. They allow modular queries

D. All of the above

Answer: D

Explanation: Subqueries in FROM can be aliased and treated as tables.

Q36. What will happen with this query?

sql

CopyEdit

SELECT department, AVG(salary) FROM employees HAVING AVG(salary) > 60000;

A. Error because GROUP BY is missing

B. Works fine

C. Filters rows

D. Returns all departments

Answer: A

Explanation: HAVING requires GROUP BY to work properly.

Q37. What is the difference between WHERE and HAVING in subqueries?

- A. WHERE filters groups, HAVING filters rows
- B. WHERE filters rows before grouping, HAVING filters groups after grouping
- C. No difference
- D. Both filter after grouping

Answer: B

Explanation: WHERE filters rows before aggregation; HAVING filters groups.

Q38. How do you write a query to find the second highest salary using subquery?

A.

sql

CopyEdit

```
SELECT MAX(salary) FROM employees WHERE salary < (SELECT MAX(salary) FROM employees);
```

B.

sql

CopyEdit

SELECT salary FROM employees ORDER BY salary DESC LIMIT 1 OFFSET 1;

- C. Both are valid
- D. Neither is valid

Answer: C

Explanation: Both queries get the second highest salary.

Q39. Which of the following can cause performance issues with subqueries?

- A. Correlated subqueries
- B. Simple SELECTs
- C. Aggregate functions
- D. Aliasing

Answer: A

Explanation: Correlated subqueries can be slow as they execute repeatedly.

Q40. Which of these is the best practice when using aggregate functions on large datasets?

- A. Avoid indexes
- B. Use indexes on grouping columns
- C. Use SELECT * always
- D. Use subqueries in WHERE clause

Answer: B

Explanation: Indexes on grouping columns improve aggregation performance.